

Smart Choices for Measurement Models

Executable Manuscript Tutorial for your Confirmatory Factor Analysis in
R Environment

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Abstract

This article aims to accomplish three objectives: first, to compile guidelines for the application of Confirmatory Factor Analysis (CFA), a widely utilized technique in applied social sciences; second, to demonstrate how these guidelines can be practically implemented through a real-world example; and third, to structure this narrative using tools that promote reproducibility, replicability, and transparency of results. To this end, we propose a solution in the form of a tutorial article wherein the key decisions made in conducting a CFA are validated through recent literature and presented within a dynamic document framework. This framework enables readers to access the article's source code, utilized data, analytical execution codes, and various reading media. We anticipate that by employing this pedagogical approach, developed entirely within an open environment (<https://phdpablo.github.io/smart-cfa/>), researchers proficient in specific statistical techniques relevant to their domains will adopt and disseminate this proposal, thereby benefiting their colleagues.

Keywords: Confirmatory Factor Analysis, Structural Equation Modeling, Internal Structure Validity, Open Science, *lavaan*.

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1 Introduction

Confirmatory Factor Analysis (CFA) is a key method for assessing the validity of a measurement instrument through its internal structure (Bandalos, 2018; Hughes, 2018; Sireci & Sukin, 2013). Validity is arguably the most crucial characteristic of a measurement model (Furr, 2021), as it addresses the essential question of what measuring instruments truly assess (Bandalos, 2018). This concern is closely linked with the classical definition of validity: the degree to which a test measures what it claims to measure (Bandalos, 2018; Furr, 2021; Sireci & Sukin, 2013; Urbina, 2014), aligning with the tripartite model still embraced by numerous scholars (Widodo, 2018).

The tripartite model of validity frames the concept using three categories of evidence: content, criterion, and construct (Bandalos, 2018). Content validity pertains to the adequacy and representativeness of test items relative to the domain or objective under investigation (Cohen et al., 2022). Criterion validity is the correlation between test outcomes and a significant external criterion, such as performance on another measure or future occurrences (Cohen et al., 2022). Construct validity evaluates the test's capacity to measure the theoretical construct it is intended to assess, taking into account related hypotheses and empirical data (Cohen et al., 2022).

Introduced in the American Psychological Association (APA) "Standards for Educational and Psychological Testing" in 1966, the tripartite concept of validity has been a cornerstone in the social sciences for decades (Bandalos, 2018). However, its fragmented and confusing nature has led to widespread criticism, prompting a shift towards a more holistic view of validity (Sireci & Sukin, 2013). This evolution was signified by the publication of the 1999 standards (AERA et al., 1999), and further by the 2014 standards (AERA et al., 2014), which redefined test validity in terms of the interpretations and uses of test scores (Furr, 2021). Under this new paradigm, validation requires diverse theoretical and empirical evidence, recognizing validity as a unified concept – construct validity – encompassing various evidence sources for evaluating potential interpretations of test scores for specific purposes (Furr, 2021; Urbina, 2014).

Thus, key authorities in psychological assessment now define validity as the degree to which evidence and theory support the interpretations of test scores for their intended purposes (AERA et al., 2014). Validity involves a comprehensive evaluation of how well empirical evidence and theoretical rationales uphold the conclusions and actions derived from test scores or other assessment types (Bandalos, 2018; Furr, 2021; Urbina, 2014).

According to APA guidelines (AERA et al., 2014), five types of validity evidence are critical: content, response process, association with external variables, consequences of test use, and internal structure. Content validity examines the extent to which test content accurately represents the domain of interest exclusively (Furr, 2021). The response process refers to the link between the construct and the specifics of the examinees' responses (Sireci & Sukin, 2013). Validity based on external variables concerns the test's correlation with other measures or constructs expected to be related or unrelated to the evaluated construct (Furr, 2021). The implications of test use focus on the positive or negative effects on the individuals or groups assessed (Bandalos, 2018).

Evidence based on internal structure assesses how well the interactions among test items and their components align with the theoretical framework used to explain the outcomes of the measurement instrument (AERA et al., 2014; Rios & Wells, 2014). Sources of internal structural validity evidence may include analyses of reliability, dimensionality, and measurement invariance.

Reliability is gauged by internal consistency, reflecting i) the reproducibility of test scores under consistent conditions and ii) the ratio of true score variance to observed score variance (Rios & Wells, 2014). Dimensionality analysis aims to verify if item interrelations support the inferences made by the measurement model's scores, which are assumed to be unidimensional (Rios & Wells, 2014). Measurement invariance confirms that item properties remain consistent across specified groups, such as gender or ethnicity.

CFA facilitates the integration of these diverse sources to substantiate the validity of the internal structure (Bandalos, 2018; Flora & Flake, 2017; Hughes, 2018; Reeves &

Marbach-Ad, 2016; Rios & Wells, 2014). In the applied social sciences, researchers often have a theoretical dimensional structure in mind (Sireci & Sukin, 2013), and CFA is employed to align the structure of the hypothesized measurement model with the observed data (Rios & Wells, 2014).

CFA constitutes a fundamental aspect of the covariance-based Structural Equation Modeling (SEM) framework (CB-SEM) (Brown, 2023; Harrington, 2009; Jackson et al., 2009; Kline, 2023; Nye, 2022). SEM is a prevalent statistical approach in the applied social sciences (Hoyle, 2023; Kline, 2023), serving as a generalization of multiple regression and factor analysis (Hoyle, 2023). This methodology facilitates the examination of complex relationships between variables and the consideration of measurement error, aligning with the requirements for measurement model validation (Hoyle, 2023).

Applications of CFA present significant complexities (Crede & Harms, 2019; Flake et al., 2017; Flake & Fried, 2020; Jackson et al., 2009; Nye, 2022; Rogers, 2024), influenced by data structure, measurement level of items, research goals, and other factors. CFA can proceed smoothly in scenarios involving unidimensional measurement models with continuous items and large samples, but may encounter challenges, such as diminished SEM flexibility, when dealing with multidimensional models with ordinal items and small sample sizes (Rogers, 2024).

This leads to an important question: Can certain strategies within CFA applications simplify the process for social scientists seeking evidence of validity in the internal structure of a measurement model? This inquiry does not suggest that research objectives should conform to quantitative methods. Rather, research aims guide scientific inquiry, defining our learning targets and priorities. Quantitative methods serve as tools towards these ends, not as objectives themselves. They represent one among many tools available to researchers, with the study’s purpose dictating method selection (Pilcher & Cortazzi, 2023).

However, as the scientific method is an ongoing journey of discovery, many questions, especially in Psychometrics concerning measurement model validation, remain open-ended. The lack of consensus on complex and varied topics suggests researchers should opt for paths offering maximal analytical flexibility, enabling exploration of diverse methodologies and solutions while keeping research objectives forefront (Price, 2017).

A recurrent topic in Factor Analysis (FA) is how to handle the measurement level of scale items. Empirical studies (Rhemtulla et al., 2012; Robitzsch, 2020, 2022) advocating for the treatment of scales with five or more response options as continuous variables have shown to enhance CFA flexibility and address validity evidence for the internal structure. The FA literature acknowledges methodological dilemmas faced when dealing with binary and/or ordinal response items with fewer than five options (Rogers, 2022, 2024).

For continuous scale items, the maximum likelihood (ML) estimator and its robust variations are applicable. For non-continuous items, estimators from the Least Squares (cat-LS) family are recommended (Nye, 2022; Rogers, 2022, 2024). Though cat-LS estimators impose fewer assumptions on data, they require larger sample sizes, more computational power, and greater researcher expertise (Robitzsch, 2020).

Assessing model fit is more challenging with cat-LS estimated models compared to those estimated by ML, which are better established and more familiar to researchers (Rhemtulla et al., 2012). Despite their increasing popularity, cat-LS models are newer,

less recognized, and seldom available in software (Rhemtulla et al., 2012). Handling missing data remains straightforward with ML models using the Full Information ML (FIML) method but is problematic with ordinal data (Rogers, 2024).

Thus, we can optimize the potential of some of the available software (Arbuckle, 2019; Bentler & Wu, 2020; Fox, 2022; JASP Team, 2023; Jöreskog & Sörbom, 2022; Muthén & Muthén, 2023; Neale et al., 2016; Ringle et al., 2022; Rosseel, 2012; The jamovi project, 2023) and overcome many of the limitations for ordinal and nominal data, which are still present in some of them (Arbuckle, 2019; Bentler & Wu, 2020; Neale et al., 2016; Ringle et al., 2022).

This discussion does not intend to oversimplify, digress, or claim superiority of one software over another. Rather, it underscores a fundamental statistical principle: transitioning from nominal to ordinal and then to scalar measurement levels increases the flexibility of statistical methods. Empirical studies in CFA support these clarifications (Rhemtulla et al., 2012; Robitzsch, 2020, 2022).

This article assists applied social scientists in decision-making from selecting a measurement model to comparing and updating models for enhanced CFA flexibility. It addresses power analysis, data preprocessing, estimation procedures, and model modification from three angles: smart choices or recommended practices (Flake et al., 2017; Nye, 2022; Rogers, 2024), pitfalls to avoid (Crede & Harms, 2019; Rogers, 2024), and essential reporting elements (Flake & Fried, 2020; Jackson et al., 2009; Rogers, 2024).

The aim is to guide researchers through CFA to access the underlying structure of measurement models without falling into common traps at any stage of the validation process. Early-stage decisions can preempt later limitations, while missteps may necessitate exploratory research or additional efforts in subsequent phases.

Practically, this includes an R tutorial utilizing the lavaan package (Rosseel, 2012), adhering to reproducibility, replicability, and transparency standards of the Open Science movement (Gilroy & Kaplan, 2019; Kathawalla et al., 2021; Klein et al., 2018).

Tutorial articles, following the FAIR principles (Findable, Accessible, Interoperable, and Reusable) (Wilkinson et al., 2016), play a vital role in promoting open science (Martins, 2021; Mendes-Da-Silva, 2023), by detailing significant methods or application areas in an accessible yet comprehensive manner. This encourages adherence to best practices among researchers, minimizing the impact of positive publication bias.

This tutorial is structured into three sections, beyond the introductory discussion. It includes a thorough review of CFA recommended practices, an example of real-world research application in the R ecosystem, and final considerations, following Martins (2021) format for tutorial articles. This approach, combined with workflow recommendations for reproducibility, aims to support the applied social sciences community in effectively utilizing CFA (Martins, 2021; Mendes-Da-Silva, 2023).

2 Smart Choices in CFA

This paper presents a comprehensive approach to conducting a standard CFA within the applied social sciences, following the guidelines outlined by Rogers (2024). According to Rogers (2024), a typical CFA study seeks to fit a reflective common factor model with a predefined multifactor structure, established psychometric properties, and a maximum of

five Likert-type response options. This scenario frequently occurs in research endeavors where the measurement model facilitates the examination of hypotheses derived from the structural model.

The initial phase in such research involves data preprocessing. Specifically, for categorical data, Rogers (2024) advises employing multiple imputation to handle missing data, taking into consideration the limitations posed by available software and methodologies (Rogers, 2024). When a measurement model allows for the treatment of items as continuous variables, addressing this challenge can be deferred to the estimation process stage through the selection of an appropriate estimator (Robitzsch, 2022).

This paper reinterprets the insights from Rogers (2024) for CFAs that accommodate continuous item treatment. Thus, a strategic choice involves opting for measurement models that permit this approach, thereby circumventing methodological hurdles (Robitzsch, 2020, 2022) associated with binary and/or ordinal response items with up to four or five gradations. Such a decision influences various aspects of the research process, including the choice of software, power analysis, estimation techniques, criteria for model adjustment, and model comparisons. These choices, in turn, affect requirements concerning sample size, computational resources, and the researcher’s expertise (Robitzsch, 2020).

Subsequent sections delve into themes previously summarized by Rogers (2024), specifically concerning CFAs with ordinal items. These themes are explored in terms of recommended practices (Flake et al., 2017; Nye, 2022; Rogers, 2024), pitfalls to avoid (Crede & Harms, 2019; Rogers, 2024), and reporting guidelines (Flake & Fried, 2020; Jackson et al., 2009; Rogers, 2024), all within the context of selecting measurement models that accommodate continuous data interpretation.

Assuming that readers possess a foundational understanding of the topic, this paper omits certain technical details, directing readers to authoritative texts (Brown, 2015; Kline, 2023) and scholarly articles that provide an introduction to Covariance-Based Structural Equation Modeling (CB-SEM) (Davvetas et al., 2020; Shek & Yu, 2014). The discussion is framed within the CB-SEM paradigm (Brown, 2015; Jackson et al., 2009; Kline, 2023; Nye, 2022), with a focus on CFA. The paper explicitly excludes discussions on measurement model modifications in Variance-Based SEM (VB-SEM), which are predominantly addressed in the literature on Partial Least Squares SEM (PLS-SEM) (Hair et al., 2017, 2022; Henseler, 2021).

2.1 Measurement Model Selection

Selecting an appropriate measurement model is a critical initial step in the research process. For robust analysis, it is advisable to prioritize models that provide five or more ordinal response options. Research has shown that a higher number of response gradations enhances the ability to detect inaccurately defined models (Green et al., 1997; Maydeu-Olivares et al., 2017), even when using estimators designed for ordinal items (Xia & Yang, 2018). This strategy also mitigates some of the methodological challenges associated with the analysis of ordinal data in CFA (Rhemtulla et al., 2012; Robitzsch, 2020, 2022).

When choosing a measurement scale, it is crucial to select ones that have been validated in the language of application and with the study’s target audience (Flake et al., 2017). Avoid scales that are proprietary or specific to certain professions. An examination of your country’s Psychological Test Assessment System can be an effective starting

point. If the desired scale is not found within these resources, consider looking into scales developed with the support of public institutions, non-governmental organizations, research centers, or universities, as these entities often invest significant resources in validating measurement models for broader public policy purposes.

An extensive literature review is essential for selecting a suitable measurement model. This should include consulting specialized journals, books, technical reports, and academic dissertations or theses. Schumacker et al. (2021) provide a detailed guide for initiating this search. Consideration should also be given to systematic reviews or meta-analyses focusing on measurement models related to your topic of interest. It is important to review both the original articles on the scales and subsequent applications. Kline (2016) offers a useful checklist for assessing various measurement methods.

Incorporate control questions, such as requiring respondents to select “strongly agree” on specific items, and monitor survey response times to gauge participant engagement (Collier, 2020).

Avoid adopting measurement models designed for narrow purposes or those lacking rigorous psychometric validation (Flake & Fried, 2020; Kline, 2016). The mere existence of a scale does not ensure its validity (Flake et al., 2017). Also, steer clear of seldom-used or outdated scales, as they may have compromised psychometric properties. Translating a scale from another language for immediate use without thorough translation and retranslation processes is inadvisable. Be cautious of overlooking alternative factorial structures (e.g., higher-order or bifactor models) that could potentially salvage the research if considered thoroughly (Crede & Harms, 2019).

When selecting a scale, justify its choice by highlighting its strong psychometric properties, including previous empirical evidence of its application within the target population and its reliability and validity metrics (Flake & Fried, 2020; Jackson et al., 2009; Kline, 2016). If the scale has multiple potential factorial structures, provide a rationale for the chosen model to prevent the misuse of CFA for exploratory purposes (Jackson et al., 2009).

Clearly specify the selected model and rationalize your choice by detailing its advantages over other theoretical models. Illustrating the models under consideration can further clarify your research approach (Jackson et al., 2009). Finally, identify and explain any potential cross-loadings based on prior empirical evidence (Brown, 2023; Nye, 2022), ensuring a comprehensive and well-justified methodological foundation for your study.

2.2 Power Analysis

When addressing Power Analysis (PA) in CFA and SEM, it’s essential to move beyond general rules of thumb for determining sample sizes. Commonly cited guidelines suggesting minimum sizes or specific ratios of observations to parameters (e.g., 50, 100, 200, 300, 400, 500, 1000 for sample sizes or 20/1, 10/1, 5/1 for observation/parameter ratios) (Kline, 2023; Kyriazos, 2018) are based on controlled conditions that may not directly transfer to your study’s context.

Reliance on lower-bound sample sizes as a substitute for thorough PA risks inadequate power for detecting meaningful effects in your model (Wang, 2023; Westland, 2010). Tools like Soper’s calculator (<https://www.danielsoper.com/statcalc/>), while popular and frequently cited (as of 02/20/2024, with almost four years of existence, it had collected more than 1,000 citations on Google Scholar), should not replace a tailored

PA approach. Such calculators, despite their utility, may not fully accommodate the complexities and specific requirements of your research design (Feng & Hancock, 2023; Kyriazos, 2018; Moshagen & Bader, 2023).

A modern perspective on sample size determination emphasizes customizing power calculations to fit the unique aspects of each study, incorporating specific research settings and questions (Feng & Hancock, 2023; Moshagen & Bader, 2023). This approach underscores that there is no universal sample size or minimum that applies across all research scenarios (Kline, 2023).

Planning for PA should ideally precede data collection, enhancing the researcher's understanding of the study and facilitating informed decisions regarding the measurement model based on existing literature and known population characteristics (Feng & Hancock, 2023; Leite et al., 2023). A priori PA not only ensures adequate sample size for detecting the intended effects, minimizing Type II errors, but also aids in budgeting for data collection and enhancing overall research design (Feng & Hancock, 2023).

PA in SEM can be approached analytically, using asymptotic theory, or through simulation methods. Analytical methods require specifying the effect size in relation to the non-centrality parameter, while simulated PA leverages a population model to empirically estimate power (Feng & Hancock, 2023; Moshagen & Bader, 2023). These approaches are applicable to assessing both global model fit and specific model parameters.

For CFA, evaluating the power related to the global fit of the measurement model is recommended (Nye, 2022). Although analytical solutions have their limitations, they can serve as preliminary steps, complemented by simulation techniques for a more comprehensive PA (Feng & Hancock, 2023; Moshagen & Bader, 2023).

Several resources offer analytical solutions for global fit PA, including ShinyApps by Jak et al. (2021), Moshagen & Bader (2023), Wang & Rhemtulla (2021), and Zhang & Yuan (2018), with the last application providing a comprehensive suite for Monte Carlo Simulation (SMC) that accommodates missing data, non-normal distributions, and facilitates model testing without extensive coding (Wang & Rhemtulla, 2021). For an overview of these solutions and a discussion of analytical approaches, see Feng & Hancock (2023), Jak et al. (2021), Nye (2022), and Wang (2023).

However, it is a smart decision to run an SMC for the PA of your CFA model using solutions that are consistent with the results' reproducibility and replicability. In this way, even analytical solutions that the researcher may use as a starting point are recommended in the R environment via the `semTools` packages (Jak et al., 2021) and `semPower 2` (Jobst et al., 2023; Moshagen & Bader, 2023). The first option is compatible with the `lavaan` syntax and looks to be enough. The second, albeit including SMC in some cases, has a more difficult syntax.

For detailed and tailored PA, especially in complex models or unique study designs, the `simsem` package offers a robust solution, allowing for the relaxation of traditional assumptions and supporting the use of robust estimators. This package, which utilizes the familiar `lavaan` syntax, simplifies the learning curve for researchers already accustomed to SEM analyses, providing a user-friendly interface for conducting SMC (Pornprasertmanit et al., 2022).

Publishing the sampling design and methodology enhances the reproducibility and replicability of research, contributing to the scientific community's collective understanding

and validation of measurement models (Flake et al., 2017; Flake et al., 2022; Flake & Fried, 2020; Leite et al., 2023). In the context of CFA, acknowledging the power limitations of your study can signal potential concerns for the broader inferences drawn from your research, emphasizing the importance of external validity and the relevance of the outcomes over mere precision (Leite et al., 2023).

2.3 Pre-processing

Upon gathering and tabulating original data, ideally in non-binary formats such as CSV, TXT, or JSON, the first step in data preprocessing should be to eliminate responses from participants who have abandoned the study. This identification often occurs at the end of preprocessing, where these incomplete responses can offer insights into handling missing data, outliers, and multicollinearity.

Incorporating control questions and measuring response time allows researchers to further refine their dataset by excluding participants who fail control items or complete the survey unusually quickly (Collier, 2020). Calculating individual response variability (standard deviation) can identify respondents who may not have engaged meaningfully with the survey, indicated by minimal variation in their responses.

These preliminary data cleaning steps are fundamental yet frequently overlooked in empirical research. They can significantly enhance data quality before engaging in more complex statistical analyses. Visual and descriptive examination of measurement model items is implicitly beneficial for any statistical investigation and should be considered standard practice.

While data transformation methods like linearization or normalization are available, they are generally not necessary given the robust estimation processes that can handle non-normal data (Brown, 2015). Parceling items is also discouraged due to its potential to obscure underlying multidimensional structures (Brown, 2015; Crede & Harms, 2019).

Addressing missing data, outliers, and multicollinearity is critical. Single imputation methods should be avoided as they underestimate error variance and can lead to identification problems in your model (Enders, 2023). For missing data under 5%, the impact may be minimal, but for higher rates, Full Information ML (FIML) or Multiple Imputation (MI) should be utilized, with FIML often being the most straightforward and effective choice for CFA (Brown, 2015; Kline, 2023).

FIML and MI are preferred for handling missing data due to their ability to produce consistent and efficient parameter estimates under conditions similar to MI (Enders, 2023; Kline, 2023). FIML it can be adapted for non-normal data using robust estimators (Brown, 2015).

Calculating the Variance Inflation Factor (VIF) helps identify items with problematic multicollinearity ($VIF > 10$), which should be addressed to prevent model convergence issues and misinterpretations (Kline, 2016; Whittaker & Schumacker, 2022). Reflective constructs in CFA require some level of item correlation but not to the extent that it causes statistical or validity concerns.

Consider multivariate outliers rather than univariate ones, identifying and assessing their exclusion based on sample characteristics. Reporting all data cleaning processes, including any loss of items and strategies for assessing respondent engagement, is crucial for transparency. Additionally, documenting signs of multicollinearity and the software

or packages used (with versions) enhances the reproducibility and credibility of the research (Flake & Fried, 2020; Jackson et al., 2009).

Finally, making raw data public adheres to the principles of open science, promoting transparency and allowing for independent validation of research findings (Crede & Harms, 2019; Flake et al., 2022; Flake & Fried, 2020). This practice not only contributes to the scientific community’s collective knowledge base but also reinforces the integrity and reliability of the research conducted.

2.4 Estimation Process

In CFA with ordinal items, such as those involving Likert-type scales with up to five points, Rogers (2024) advocates for the use of estimators from the Ordinary Least Squares (OLS) family. Specifically, for smaller samples, the recommendation is to utilize the Unweighted Least Squares (ULS) in its robust form (RULS), and for larger samples, the Diagonally Weighted Least Squares (DWLS) in its robust version (RDWLS), citing substantial supporting research.

Despite this, empirical evidence (Rhemtulla et al., 2012; Robitzsch, 2022) and theoretical considerations (Robitzsch, 2020) suggest that treating ordinal data as continuous can yield acceptable outcomes when the response options number five or more. Particularly with 6-7 categories, comparisons between methods under various conditions reveal little difference, and it is recommended to use a greater number of response alternatives (5) to enhance the power for detecting model misspecifications (Maydeu-Olivares et al., 2017).

The ML estimator, noted for its robustness to minor deviations from normality (Brown, 2015), is further improved by using robust versions like MLR (employing Huber-White standard errors and Yuan-Bentler scaled χ^2). This adjustment allows for generating robust standard errors and adjusted test statistics, with MLR offering extensive applicability including in scenarios of missing data (RFIML) or where data breaches the independence of observations assumption (Brown, 2015; Rosseel, 2012). Comparative empirical studies have supported the effectiveness of MLR against alternative estimators (Bandalos, 2014; Holgado-Tello et al., 2016; Li, 2016; Nalbantoğlu-Yılmaz, 2019; Yang & Liang, 2013; Yang-Wallentin et al., 2010).

Researchers are advised to carefully describe and justify the chosen estimation method based on the data characteristics and the specific model being evaluated (Crede & Harms, 2019). It is also critical to report any estimation challenges encountered, such as algorithm non-convergence or model misidentification (Nye, 2022). In case of estimation difficulties, alternative approaches like MLM estimators (employing robust standard errors and Satorra-Bentler scaled χ^2) or the default ML with non-parametric bootstrapping, as proposed by Bollen-Stine, can be considered. This latter approach is also capable of accommodating missing data (Brown, 2015; Kline, 2023).

Additionally, it is important to clarify whether the variance of a marker variable was fixed (=1) to scale the latent variables (Jackson et al., 2009), and to provide both standardized and unstandardized parameter estimates (Nye, 2022). These steps are crucial for ensuring transparency, reproducibility, and the ability to critically assess the validity of the CFA results.

2.5 Model Fit

In conducting CFA with ordinal items, such as Likert-type scales, it's crucial to approach model evaluation with nuance and avoid reliance on rigid cutoff values for fit indices. Adhering strictly to traditional cutoffs – whether more conservative (e.g., SRMR .06, RMSEA .06, CFI .95) or less conservative (e.g., RMSEA .08, CFI .90, SRMR .08) – should not be the sole criterion for model acceptance (Xia & Yang, 2019). The origins of these thresholds are in simulation studies with specific configurations (up to three factors, fifteen items, factor loadings between 0.7 and 0.8) (West et al., 2023), and may not universally apply due to the variance in the number of items, factors, model degrees of freedom, misfit types, and presence of missing data (Groskurth et al., 2023; Niemand & Mai, 2018; West et al., 2023).

Evaluation of global fit indices (SRMR, RMSEA, CFI) should be done in a collective manner, rather than fixating on any single index. A deviation from traditional cutoffs warrants further investigation into whether the discrepancy is attributable to data characteristics or limitations of the index, rather than indicating a fundamental model misspecification (Nye, 2022). Interpreting fit indices as effect sizes can offer a more meaningful assessment of model fit, aligning with their original conceptualization (McNeish, 2023b; McNeish & Wolf, 2023b).

The SRMR is noted for its robustness across various conditions, including non-normality and different measurement levels of items. Pairing SRMR with CFI can help balance Type I and Type II errors, but reliance on alternative indices may increase the risk of Type I error (Mai et al., 2021; Niemand & Mai, 2018).

Emerging methods like the Dynamic Fit Index (DFI) and Flexible Cutoffs (FCO) offer tailored approaches to evaluating global fit. DFI, based on simulation, provides model-specific cutoff points, adjusting simulations to match the empirical model's characteristics (McNeish, 2023a; McNeish & Wolf, 2023a; McNeish & Wolf, 2023b). FCO, while not requiring identification of a misspecified model like DFI, conservatively defines misfit, shifting focus from approximate to accurate fit (McNeish & Wolf, 2023a).

For those hesitant to delve into simulation-based methods, Equivalence Testing (EQT) presents an alternative. EQT aligns with the analytical mindset of PA and incorporates DFI principles, challenging the conventional hypothesis testing framework by considering model specification and misspecification size control (Yuan et al., 2016).

When addressing reliability, Cronbach's Alpha should not be the default measure due to its limitations. Instead, consider McDonald's Omega or the Greatest Lower Bound (GLB) for a more accurate reliability assessment within the CFA context (Bell et al., 2023; Cho, 2022; Dunn et al., 2014; Flora, 2020; Goodboy & Martin, 2020; Green & Yang, 2015; Hayes & Coutts, 2020; Kalkbrenner, 2023; McNeish, 2018; Trizano-Hermosilla & Alvarado, 2016).

Before modifying the model, first check for Heywood instances, which are standardized factor loadings greater than one or negative variances (Nye, 2022) and document the chosen cutoffs for evaluation. Tools and resources like ShinyApp for DFI and the FCO package in R can facilitate the application of these advanced methodologies (Mai et al., 2021; McNeish & Wolf, 2023b; Niemand & Mai, 2018). Always report corrected chi-square and degrees of freedom, alongside a minimum of three global fit indices (RMSEA, CFI, SRMR) and local fit measures to provide a comprehensive view of model fit and adjustment decisions (Crede & Harms, 2019; Flake & Fried, 2020).

2.6 Model Comparisons and Modifications

Researchers embarking on CFA should avoid prematurely committing to a specific factor structure without thoroughly evaluating and comparing alternate configurations. It's advisable to consider various potential structures early in the study design, ensuring the selected model is based on its merits relative to competing theories (Jackson et al., 2009). Since models are inherently approximations of reality, adopting the most effective "working hypothesis" is a dynamic process, contingent on ongoing assessments against emerging alternatives (Preacher & Yaremych, 2023).

Good models are characterized not only by their interpretability, simplicity, and generalizability but notably by their capacity to surpass competing models in critical aspects. This competitive advantage frames the selected theory as the prevailing hypothesis until a more compelling alternative is identified (Preacher & Yaremych, 2023).

The evaluation of model fit should extend beyond isolated assessments using fit indices. A comprehensive approach involves comparing multiple models, each grounded in substantiated theories, to discern the most accurate representation of the underlying structure. This comparative analysis is preferred over singular model evaluations, fostering a more holistic understanding of the phenomena under study (Preacher & Yaremych, 2023).

Uniform application of models across the same dataset, utilizing identical software and sample size, ensures consistency in the researcher's analytical freedom, mitigating the risk of results manipulation. This standardized approach underpins a more rigorous and transparent investigative process (Preacher & Yaremych, 2023).

Model selection is instrumental in pinpointing the most effective explanatory framework for the observed phenomena, enabling the dismissal of less performance models while retaining promising ones for further exploration. This methodological flexibility enhances the depth of analysis, contributing to the advancement of knowledge within the social sciences (Preacher & Yaremych, 2023).

Adjustments to a model, particularly in response to unsatisfactory fit indices, should be theoretically grounded and reflective of findings from prior research. Blind adherence to a pre-established model or making hasty modifications can adversely affect the structural model's integrity. Thoughtful adjustments, potentially revisiting exploratory factor analysis (EFA) or considering Exploratory SEM (ESEM) for cross-loadings representation, are preferable to drastic changes that might shift the study from confirmatory to exploratory research (Brown, 2023; Crede & Harms, 2019; Flake et al., 2017; Jackson et al., 2009).

All modifications to the measurement model, especially those enhancing model fit, must be meticulously documented to maintain transparency and support reproducibility (Flake & Fried, 2020). Openly reporting these adjustments, including item exclusions and inter-item correlations, is vital for the scientific integrity of the research (Flake et al., 2022; Nye, 2022).

Regarding model comparison and selection, traditional fit indices (SRMR, RMSEA, CFI) have limitations for direct model comparisons. Adjusted chi-square tests and information criteria like AIC and BIC are more suitable for this purpose, balancing model fit and parsimony. These criteria, however, should be applied with an understanding of their constraints and complemented by theoretical judgment to inform model selection decisions (Brown, 2015; Huang, 2017; Lai, 2020, 2021; Preacher & Yaremych, 2023).

Ultimately, model selection in SEM is a nuanced process, blending empirical evidence with theoretical insights. Researchers are encouraged to leverage a range of models based on theoretical foundations, ensuring that the eventual model selection is not solely determined by statistical criteria but is also informed by substantive theory and expertise (Preacher & Yaremych, 2023). This balanced approach underscores the importance of theory-driven research in the social sciences, guiding the interpretation and application of findings derived from chosen models.

3 Executable Manuscript

3.1 Measurement Model Selection

3.2 Power Analysis

3.3 Pre-processing

3.4 Estimation Process

3.5 Model Fit

3.6 Model Comparisons and Modifications

4 Final Considerations

5 Conclusion

References

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